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Data-driven surrogate optimization for deploying heterogeneous multi-energy storage to improve demand response performance at building cluster level

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HIGHLIGHTS

- Accurate and explicit surrogate model by symbolic regression assists optimization.
- Most suitable energy storage configurations identified for respective buildings.
- Improve demand response by cost-effective budget allocation at a cluster level.
- Optimally match energy use pattern with heterogeneity of multi-energy storage.

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ABSTRACT

Energy storage such as battery and thermal energy storage is an effective approach to shift building peak load and alleviate grid stress at a building cluster level. However, due to the heterogeneous performance of different types of storage (e.g., response speed, charge/discharge efficiency and rate, storage capacity) and highly diversified energy use patterns of individual buildings, the multi-energy storage should be properly selected and optimally designed for individual buildings to achieve effective load shifting. The optimal deployment of multi-energy storage at a cluster level is a challenging optimization problem due to the nonlinear dynamic performance of the multi-energy storage and the high dimensionality as a result of a large number of buildings. To tackle the challenges, this study proposes a data-driven surrogate optimization method that optimally deploys multi-energy storage at a cluster level to minimize the building cluster energy bill under demand response programs. The method utilizes data-driven surrogate models to accurately predict demand response performance of individual buildings with multi-energy storage. An iterative optimization with automated energy-storage-option screening is developed to optimize the multi-energy storage configurations and design parameters. For a case study including 21 buildings, by optimally deploying multi-energy storage including battery, cooling TES tank, and building-integrated TES, the method reduced the building cluster energy bill by 8%–181% as compared to baseline cases. The optimal deployment method effectively identifies the buildings with better potential to adopt demand-side management and balances the pros and cons of the energy storage options, increasing demand response incentives by 12%–31%. The proposed method can be used in practice to facilitate the deployment of energy storage and improve engagement of buildings in demand response.

1. Introduction

Building is one of the most energy-intensive sectors worldwide [1]. In a metropolis such as Hong Kong, the building sector accounts for ~90% of the electricity consumption [2]. The escalating building electrical demand is causing unmanageable stress on the power grid, and failing to meet the demand may cause cascade grid failures and blackouts [3,4]. Demand-side management such as load shifting, and peak

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shaving is an effective approach to reducing peak loads of buildings and alleviating the grid stress [5,6]. Many power suppliers are providing demand response programs to urge users to adopt demand-side management measures [7,8].

Effective use of energy storage can significantly improve the demand response performance and energy flexibility of buildings, thereby alleviating grid stress [9]. Based on the type of energy, there are mainly two categories of energy storage for buildings: 1) Electrical energy storage (EES) which is usually the battery; and 2) Thermal energy storage (TES) mainly including active TES tank and passive building-integrated thermal energy storage (BITES) [10]. The performance of different types of energy storage is heterogeneous in many aspects including energy type, charge/discharge rate, storage capacity, and unit cost. Meanwhile, the energy use patterns of different individual buildings are highly diversified. Selecting suitable energy storage or adopting multi-energy storage, and properly sizing the storage is critical to the demand response performance and cost-effectiveness of the building energy systems.

Extensive studies have investigated the optimal design and sizing of energy storage for buildings to improve demand response performance. The major studies focused on single buildings. Simplified approaches such as analytical method [11] and parametric analysis [12,13] have been adopted to solve energy storage sizing problems. For instance, Brandt et al. [12] optimally sized a multi-energy storage system including a battery and a TES tank via parametric analysis. An idealized dispatch approach based on simplified energy storage models was adopted to evaluate the load-shifting performance. The simplified approaches are easy to implement, but they cannot accurately characterize the dynamic performance of energy storage. Meanwhile, they cannot handle complex nonlinear optimization problems with multiple variables.

Meta-heuristic algorithms such as genetic algorithm (GA) [14,15] and particle swarm optimization (PSO) [16] were frequently used to solve complex optimal design/sizing problems of energy storage based on detailed dynamic simulations. For instance, Cui et al. [15] developed a model-based optimal design method of TES tanks with phase change material (PCM) to maximize cost savings through demand response. A GA-based technique was developed to identify the optimal design parameters of the TES tank. Baniasadi et al. [16] utilized the PSO to optimally size multi-energy storage including batteries and water tanks. A dual-loop optimization framework was developed to minimize operation costs under time-of-use (TOU) tariff and sizing the energy storage simultaneously. The meta-heuristic algorithms well handled the nonlinearity in the optimal design/sizing problems. However, these algorithms may cause unmanageable computational costs when facing high-dimensional problems due to exponentially expanded search space and numerous evaluations of the objective function (i.e., simulation runs).

Although extensive studies focused on energy storage deployment for single buildings, more recent studies have proven that the demand response performance of buildings at a cluster level rather than an individual building is the main concern for grid power balance [17,18]. Optimally deploying the EES and/or TES energy storage at a building cluster level to improve the demand response performance could more effectively alleviate the grid stress.

Several solutions were developed for the optimal deployment of energy storage for building clusters. Thanks to its effectiveness to solve high dimensional problems, linear programming optimization techniques were frequently used to solve such optimal deployment problems. Novoa et al. [19] developed a mixed integer linear programming optimal deployment method of distributed batteries and photovoltaics (PVs) in a microgrid, to minimize the overall cost. To handle the discontinuity of adopting PV/battery or not, the PV/battery adoption at individual buildings was indicated by binary integer variables. Li et al. [20] considered the multi-energy storage including batteries and hot water tanks. The optimal deployment was also formulated as a mixed-inter linear programming problem based on linear models of building thermal load, hot water tanks, and batteries. Despite its high efficiency, the linear programming-based optimization methods usually utilize explicit, and linear models. Such linear models may not accurately characterize heterogeneous multi-energy storage with diversified performance. For instance, the linear models may not effectively evaluate the performance of the BITES under demand response control strategies such as pre-cooling and zone temperature reset [21,22], since the building thermal performance is generally nonlinear [23,24].

Some studies adopted an aggregable capacity framework to optimally size the batteries for building clusters. For instance, Huang et al. [25] developed a hierarchical design method of distributed battery energy storage in communities with shared PV power. The method considered the distributed batteries as a single aggregable virtual battery, thereby simplifying the sizing problem. The aggregable capacity framework was also adopted by other studies [26,27] to solve the sizing of distributed PVs and/or batteries in energy-sharing communities. This method effectively handles the high dimensionality by considering the storage capacity aggregable. This optimization framework may not be applicable for the optimal deployment of heterogeneous multi-energy storage, since the aggregation and sharing of thermal energy storage cannot be easily achieved at a cluster level.

As an emerging technique, data-driven surrogate models have been extensively adopted in the modeling of building energy systems due to their ability to capture complex nonlinear relationships from historical data [28]. The surrogate models can approximate expensive objective functions with a limited number of function evaluations [29]. Among extensive data-driven modeling techniques, artificial neural networks [30], support vector machine/regression [31], and decision tree-based techniques [32], have been frequently adopted for surrogate modeling of building energy systems. However, most existing methods focused on modeling a single type of energy storage, and fine-tuning of model structures and hyperparameters were applied individually to improve model prediction performance. In optimal deployment of multi-energy storage at a building cluster level, a flexible data-driven surrogate framework that can automatically adapt itself to accurately characterize the demand response performance of the individual buildings and the cluster is needed, considering the heterogeneous performance of multi-energy storage and the diversified energy use patterns of individual buildings, as well as their synergetic effects.

To sum up, deploying EES and/or TES can effectively improve the demand response performance of buildings, while the existing methods may not be applicable for the optimal deployment of multi-energy storage at a cluster level. The frequently used metaheuristic algorithms for single building optimization [14–16] may not be suitable for the optimal deployment at a cluster level due to high dimensionality and escalating computational costs. The linear programming optimization method [19,20] could handle high dimensional problems while it oversimplified the dynamic nonlinear performance of energy storage. The aggregable capacity framework [25–27] was developed based on the assumption of aggregable battery storage capacity, while it may not be suitable for TES. Meanwhile, only a few studies discussed the adoption and optimal configuration of multi-energy storage [12,16]. There is still a research gap between the existing studies and the optimal deployment of multi-energy storage to improve demand response performance at a building cluster level. Such optimal deployment includes two key problems, i.e., identifying optimal energy storage configurations for individual buildings, and optimal sizing of energy storage. It should be based on accurate characterization of the energy storage dynamic performance and the diversified individual buildings' energy use patterns. Meanwhile, it needs to effectively handle high dimensional optimization, and avoid excessive runs of high-computational cost simulation.

This study, therefore, proposes a data-driven surrogate optimization method for the optimal deployment of multi-energy storage at a building cluster level to improve demand response performance. The data-driven surrogate optimization is selected for the development due to 1) the high-dimensional, complex, and nonlinear nature of the optimal

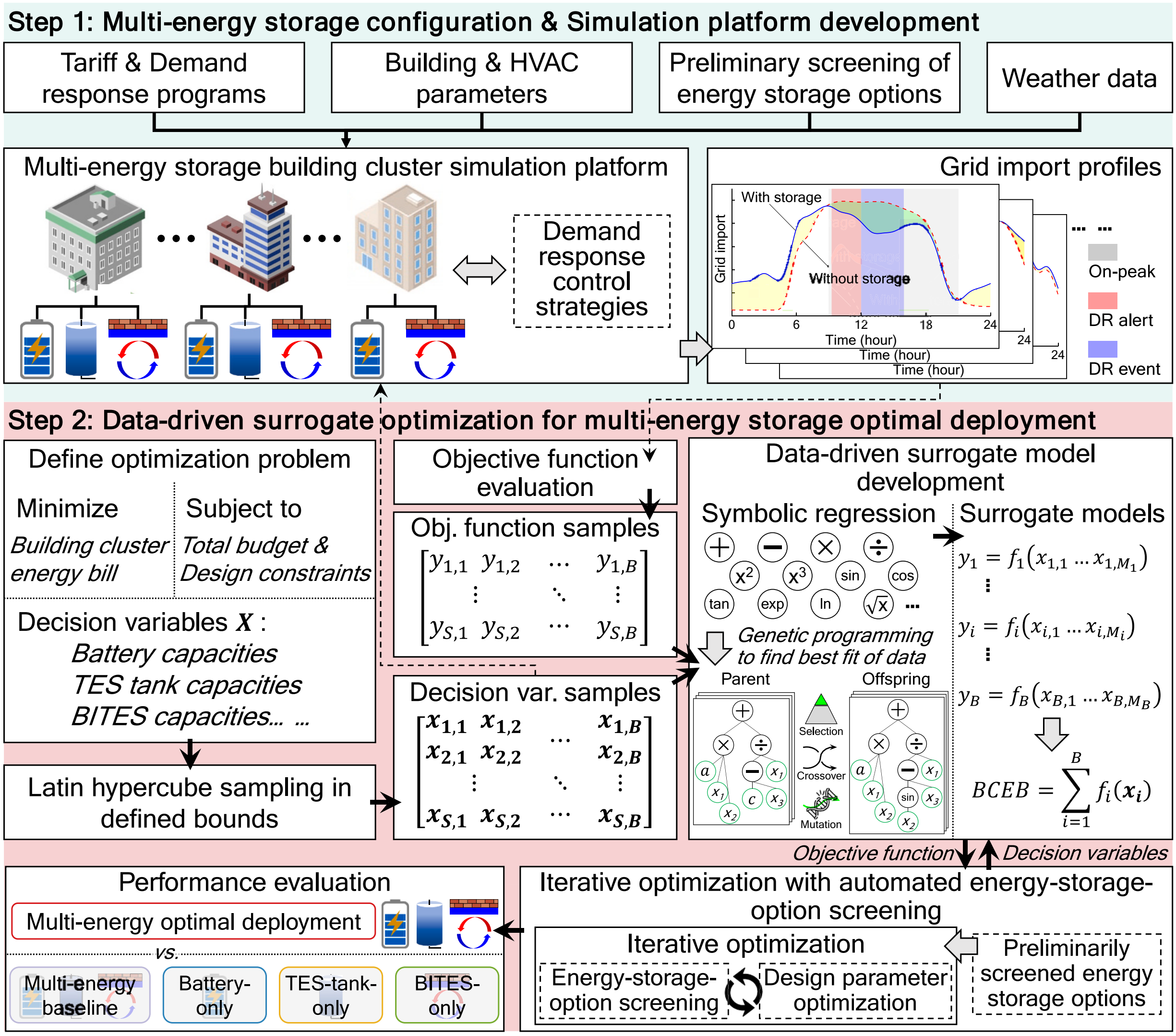


Fig. 1. Outline of the data-driven surrogate optimization for optimal deployment of multi-energy storage at a building cluster level.

deployment problem concerned; 2) the effectiveness of surrogate models to accurately approximate complex nonlinear objective functions with limited simulation runs so as to reduce computational costs; and 3) the capability of the data-driven surrogate optimization to deeply investigate the high-dimensional parameter space and identify the optimal solution. The novelty and key contribution of this study are as follows.

- Heterogeneous performance of multi-energy storage and highly diversified energy-use patterns of individual buildings are collaboratively integrated in a new data-driven surrogate optimization method to optimally deploy multi-energy storage at a building cluster level.
- A genetic programming symbolic regression technique that automatically adapts its form of equation based on training data is adopted to develop the surrogate models, accurately and efficiently characterizing diversified energy performance among individual buildings in the cluster.
- Data-driven surrogate models in conjunction with an iterative optimization with automated energy-storage-option screening are developed to effectively tackle the mixed problem involving combinatorial optimization of energy storage configuration and the high-dimensional continuous optimization of energy storage sizing.

2. Data-driven surrogate optimization method

2.1. Overall method and objective function

Fig. 1 presents the outline of the data-driven surrogate optimization method which includes two steps. In the first step, to accurately characterize the grid import of individual buildings with the multi-energy storage, and to generate training data of surrogate models, a simulation platform of the building cluster is developed. Details for the multi-energy storage configuration and the simulation platform development will be provided in Section 3. In the second step, the optimal deployment plan of the heterogeneous multi-energy storage for the building cluster is identified by using the data-driven surrogate optimization method. A genetic programming symbolic regression technique is adopted to develop explicit and high-accuracy surrogate models that predict objective function (i.e., total energy bill of the building cluster) based on decision variables (i.e., multi-energy storage configuration and corresponding design parameters). With the surrogate models, the optimal configurations and optimal design parameters of the multi-energy storage for individual buildings are identified by an iterative optimization technique, to minimize the building cluster energy bill. The effectiveness of the proposed method is evaluated by a comparative study with baseline cases of multi-energy storage and associated analysis. Details of the data-driven surrogate optimization method are described

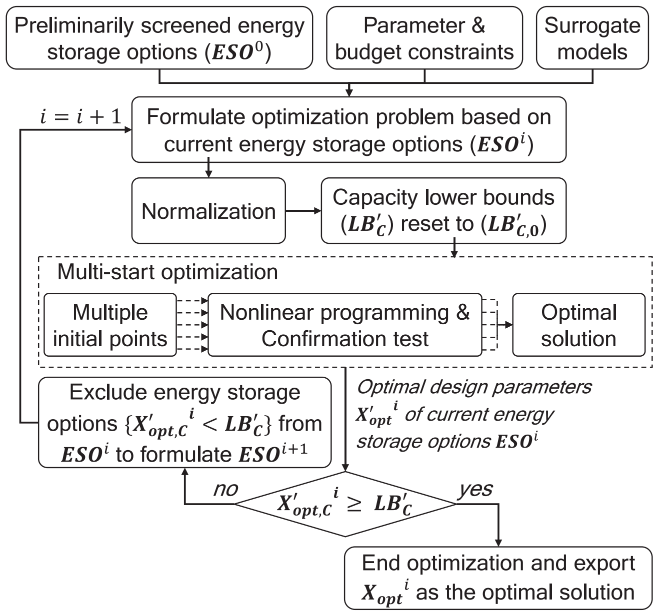


Fig. 2. Outline of the iterative optimization technique with automated energy-storage-option screening.

in the following sections.

The optimal deployment of multi-energy storage on a building cluster level is defined by Eq. (1). The objective function, i.e., building cluster energy bill, is a sum of the energy bills of individual buildings. The individual building energy bill is considered a function of the decision variables (of the building) that represent the adopted energy storage options and the design parameters. The budget constraint limits the summation of the initial investments in multi-energy storage. Meanwhile, design parameters of the energy storage options are confined by respective design constraints, i.e., $\boldsymbol{LB}$ and $\boldsymbol{UB}$, to guarantee reasonable optimal designs. The energy storage adoption at individual buildings is represented by the capacity decision variable (X_C) that is either zero or in the range of the design constraint.

$$minimize\ BCEB = \sum_{i=1}^{B} IBEB_i(\boldsymbol{x_i}) \tag{1}$$

$$subject\ to\ \sum_{i=1}^{B} II_i(\boldsymbol{x_i}) \leq Total\ budget$$

$$\boldsymbol{x_i} = \{x_{i,1}, x_{i,2}, \ldots, x_{i,M_i}\}$$

$$\boldsymbol{LB} \leq \boldsymbol{X} \leq \boldsymbol{UB}, \boldsymbol{X} = \{\boldsymbol{x_1}, \boldsymbol{x_2}, \ldots, \boldsymbol{x_B}\}$$

$$X_C = 0\ or\ [\boldsymbol{LB}_C, \boldsymbol{UB}_C], X_C \in \boldsymbol{X}$$

where $BCEB$ indicates the building cluster energy bill, B indicates the total number of buildings in the cluster, $IBEB_i(\bullet)$ and $II_i(\bullet)$ are the individual building energy bill and the initial investment of the i^{th} building, respectively, $\boldsymbol{x_i}$, is the decision variables of the i^{th} building, $x_{i,j}$ indicates the j^{th} decision variable of the i^{th} building, M_i is the number of decision variables of the i^{th} building, $\boldsymbol{X}$ is the decision variables of the building cluster, $\boldsymbol{LB}$ and $\boldsymbol{UB}$ are the upper bounds and lower bounds of the decision variables, respectively, and X_C is the capacity decision variable.

2.2. Development of surrogate models

The objective function, i.e., building cluster energy bill, is evaluated by aggregating the bills of all buildings in the cluster. Therefore, individual surrogate models, i.e., $f_i(\boldsymbol{x_i})$, can be developed for the buildings. Then, the building cluster energy bill can be predicted by aggregating the outputs of surrogate models, i.e., $BCEB = \sum_{i=1}^{B} f_i(\boldsymbol{x_i})$. By decomposing the prediction, the input dimensions of individual models are significantly reduced as compared to developing a single surrogate model to predict the building cluster energy bill. This largely simplifies the surrogate model development.

The development of the surrogate models starts with the training data generation by using the simulation platform. For each surrogate model, samples, i.e., simulation cases, are generated in the parameter space defined by the lower and upper bounds of energy storage design parameters. The Latin hypercube sampling [33] is used to generate the samples. To facilitate the energy-storage-option screening (in Section 2.3), extra samples that exclude 1 to N-1 of the energy storage options are formulated to improve the accuracy of the surrogate models when predicting buildings with screened energy storage options. The decision variable samples are formulated and sent to the simulation platform, to generate the grid-import profiles and evaluate the objective function.

To achieve high prediction accuracy at manageable computational costs, the symbolic regression [34] is adopted to develop the surrogate models. The symbolic regression is a regression technique that explores the space of explicit mathematical expressions to find the model that best fits a given dataset, considering both prediction accuracy and model simplicity. Unlike conventional regression techniques (such as linear regression and polynomial regression) that fit parameters to equations with provided forms, the symbolic regression searches the best-fit form of the equation and the corresponding best-fit parameters simultaneously. In this study, the multigene symbolic regression [35] which is an improved version of the symbolic regression is adopted as the regression technique.

2.3. Iterative optimization with automated energy-storage-option screening

To identify the optimal configurations of energy storage options and corresponding design parameters for individual buildings at a cluster level, this study develops an iterative optimization technique with automated energy-storage-option screening. The proposed technique selects the optimal configuration of energy storage options for each building via iterative parameter screening and identifies the optimal design parameters with a multi-start nonlinear programming method. The iterative optimization can effectively solve the optimal deployment problem, by decoupling the combinatorial optimization problem of selecting optimal energy storage options and the optimization of continuous energy storage design parameters. It enables the use of high-efficiency nonlinear programming techniques for high-dimensional continuous problems.

Fig. 2 presents the outline of the iterative optimization technique. It formulates an optimization problem based on the energy storage options ($\boldsymbol{ESO}^i$), and the constraints of design parameters and budget. The initial storage options ($\boldsymbol{ESO}^0$) are determined by the preliminary screening. To facilitate the iterative energy-storage-option screening, the normalized optimization lower bounds ($\boldsymbol{LB}'_C$) of normalized capacity parameters (X'_C) such as battery capacities and TES tank sizes, are reset to zero instead of the minimum capacities. The multi-start nonlinear programming is then utilized to identify the optimal solution ($X'_{opt}{}^i$) at the current energy storage options, by using the surrogate models. The energy storage options with the optimal capacity parameters ($X'_{opt,C}{}^i$) less than the normalized minimum capacities ($\boldsymbol{LB}'_C$) are considered less cost-effective for reducing the building cluster energy bill and unreasonably undersized. They are excluded from the multi-energy storage options of corresponding buildings. The iterative optimization continuously excludes the undersized energy storage and identifies new optimal design parameters of the screened energy storage options, until

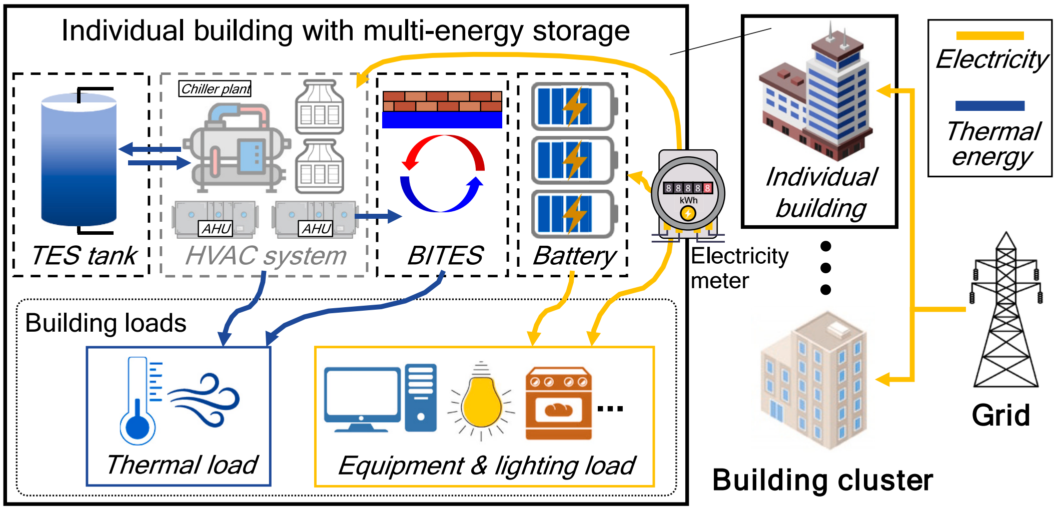


Fig. 3. Schematic of the building energy system with multi-energy storage.

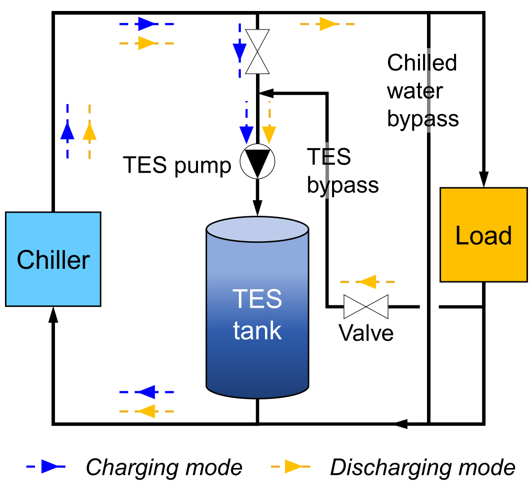


Fig. 4. Configuration and operation modes of the TES tank with PCM.

the optimal capacities of the selected energy storage options are all above or equal to the minimum capacities.

Thanks to the continuous and explicit nature of the surrogate models, a high-efficiency nonlinear programming technique, the interior-point algorithm [36], can be used to solve the optimization problem. Meanwhile, to avoid local optima, the multi-start strategy is utilized to run the interior-point nonlinear programming from multiple sets of initial decision variables and identify the optimal solution from the multiple optimization results. In this study, the interior-point nonlinear programming is conducted via the MATLAB *fmincon* function [37], and the multi-start strategy is implemented via the Latin hypercube sampling.

3. Configuration of multi-energy storage and case study setup

3.1. Energy storage options and demand response control strategies

The multi-energy storage and demand response control strategies are formulated based on the assumption that both time-of-use (TOU) tariff and incentive demand response (DR) events are provided. Fig. 3 presents the configuration of the multi-energy storage. The multi-energy storage is integrated with the building energy system via a behind-the-meter scheme to reduce grid import during on-peak hours and DR events. Three energy storage options are utilized to improve the demand response performance of the individual buildings, including the battery, the TES tank with phase change material (PCM), and the BITES with PCM. The three options are selected due to their adequate performance at reasonable costs, and because they are frequently used in current buildings and extensively investigated in existing studies [38,39]. The PCM is adopted considering its high energy density and its stable phase change temperature to store thermal energy in a certain temperature range.

The battery is dedicated to the equipment and lighting load, excluding the HVAC (heating, ventilation, and air conditioning) load. It is charged using grid power during off-peak hours. Considering the high cost and relatively small capacity of the battery, it is discharged only for DR events to reduce peak load and benefit more from the high DR incentives instead of the TOU tariff. The battery is discharged at a constant

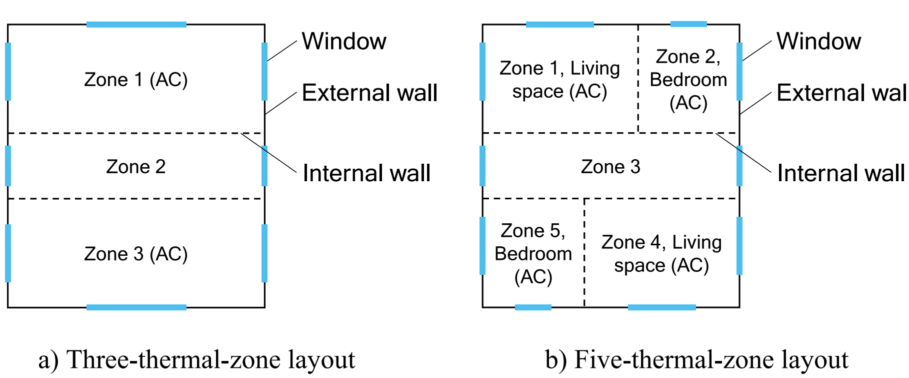


Fig. 5. Layouts of building thermal zones (where AC indicates air-conditioned).

Table 1
Building parameters of the case study building cluster.

No.	Gross area (m^2)	Perimeter (m)	Façade type[1]	WWR[2]	Thermal mass[3]	Building type	HVAC type[4]	Reset tolerance	Candidate ESOs[5]
1	23,523	288	RG	M	H	Hotel	CA-W	1.5 °C	B-T-I
2	24,322	279	GL	–	L	Hotel	CA-W	0 °C	B-T-I
3	6586	273	RG	L	M	Hotel	CA-W	1.5 °C	B-T-I
4	10,304	135	RG	M	M	Office	CA-W	1.5 °C	B-T-I
5	7208	97	GL	–	L	Office	CA-W	0 °C	B-T-I
6	4109	73	GL	–	L	Office	CA-A	0 °C	B-T-I
7	6422	98	RG	M	M	Office	CA-W	1.5 °C	B-T-I
8	9602	235	RG	M	H	Office	CA-W	1.5 °C	B-T-I
9	32,245	174	GL	–	L	Office	CA-W	0 °C	B-T-I
10	4282	85	RG	H	M	Restaurant	CA-A	1.5 °C	B-T-I
11	10,708	149	GL	–	L	Restaurant	CA-W	0 °C	B-T-I
12	1392	77	RG	M	M	Restaurant	CA-A	1.5 °C	B-T-I
13	7836	113	RG	M	M	Residential	SA	2.5 °C	B-I
14	3288	87	RG	M	M	Residential	SA	2.5 °C	B-I
15	3304	117	RG	M	M	Residential	SA	2.5 °C	B-I
16	3020	73	RG	M	M	Residential	SA	2.5 °C	B-I
17	1665	73	RG	M	M	Residential	SA	2.5 °C	B-I
18	2497	75	RG	M	M	Residential	SA	2.5 °C	B-I
19	5964	244	GL	–	L	Retail	CA-A	1.5 °C	B-T-I
20	5563	164	RG	M	M	Retail	CA-A	2.5 °C	B-T-I
21	2445	116	RG	L	M	Retail	CA-A	2.5 °C	B-T-I

[1] RG indicates regular façade and GL indicates glass curtain façade.
[2] L/M/H indicates low/medium/high-level WWR, corresponding to 0.40/0.55/0.70 WWR on external walls.
[3] L/M/H indicates low/medium/high-level thermal mass, corresponding to 0.1/0.125/0.15 m-thick concrete layer in external walls and 0.08/0.12/0.16 m-thick concrete layer in floors.
[4] CA-W indicates centralized air-conditioning system with water-cooled chiller and cooling tower, CA-A indicates centralized air-conditioning system with air-cooled chiller, and SA indicates split air-conditioner.
[5] Candidate ESOs (energy storage options) where the B-T-I indicates battery, TES tank and BITES, and B-I indicates battery and BITES.

rate that depletes the battery at the end of the DR event or at the equipment and lighting load, whichever is lower.

The TES tank is integrated with a centralized HVAC system using a downstream configuration with reference to the previous design [40], as presented in Fig. 4. The TES tank is charged in two operation modes: 1) TOU-based charging that regularly charges the TES tank during off-peak hours; and 2) DR-based charging that charges the TES tank in the hours immediately prior to DR events if the alerts are sent earlier than three hours ahead (i.e., alert lead times are more than three hours). The DR-based charging is designed to facilitate the grid-import reduction during DR events without significantly increasing the import immediately prior to the DR events. The TES tank is discharged during the on-peak hours and DR events (unless it is in the DR-based charging mode), to pre-cool a bypass of the return chilled water, thereby reducing chiller power consumption.

The BITES attaches a PCM layer at the room ceiling of air-conditioned zones, which is a passive method of integrating PCM with buildings to increase thermal mass and facilitate load shifting and peak shaving [41]. Charging and discharging of the BITES is achieved by passive heat transfer between the indoor and the PCM layer. Due to the passive configuration of the BITES, its charging and discharging cannot be directly controlled.

Apart from the multi-energy storage, two demand response control strategies, including pre-cooling and zone temperature reset, are adopted for the buildings, to shift cooling load via the BITES (if available) and other building thermal mass.

The pre-cooling is conducted based on the regular TOU schedule and also based on the irregular DR events. Regarding the TOU-based pre-cooling, it starts three hours prior to the on-peak hours till the beginning of peak hours. The DR-based precooling starts immediately after receiving the DR alert if it is sent earlier than three hours ahead. To avoid abrupt increases of HVAC power, the pre-cooling is conducted by linearly decreasing the zone temperature set point in an hour, such as from 25.5 °C to 23.5 °C, and then stay at 23.5 °C for two hours to effectively pre-cool the building thermal mass.

The zone-temperature-reset control strategy is dedicated to the DR events, which increases the zone temperature set point by a certain amount immediately after a DR event begins. Considering the different acceptable levels of thermal discomfort, the temperature increase, i.e., temperature-reset tolerance, is dependent on the building type. For instance, occupants of a high-standard office building may not accept any level of discomfort, resulting in 0 °C temperature-reset tolerance, while the temperature-reset tolerance of residential buildings could be higher, such as 2.5 °C if occupants are less sensitive to the discomfort.

3.2. Simulation platform

The simulation platform is developed to evaluate the grid import profiles and resulting energy bills of individual buildings and the building cluster. In this study, TRNSYS 17 [42] is used as the simulation platform. It mainly consists of the Type-56 building model, the battery model, the model of the TES tank with PCM, the model of the BITES with PCM, and the HVAC models.

The TRNSYS Type 56 is used to develop building models. An individual building is simulated by a typical floor for simplification. Commercial buildings such as office buildings, hotel buildings, retail buildings, and restaurant buildings, are modeled based on a three-thermal-zone layout as presented in Fig. 5a. Considering great differences between living and dining room occupancy patterns and bedroom occupancy patterns, residential buildings are modeled based on a five-thermal-zone layout as presented in Fig. 5b. In both layouts, it is assumed that 80% of the floor area is air-conditioned according to corresponding schedules.

The battery is modeled by a lumped-parameter electro-thermal model that has been extensively used in previous studies [43]. The model consists of a two resistor-capacitor equivalent-circuit electrical sub-model and a two-node thermal sub-model to accurately characterize the battery dynamic performance.

The TES tank with PCM is modeled by a semi-2D finite-difference model modified from [40] and our previous study [44,45]. The model assumes that the TES tank consists of multiple parallelly arranged identical units. Each unit is a PCM column with a pipe embedded in the center. A unit is divided into multiple nodes along the water flow direction, and in each unit, the PCM is further divided into multiple layers

Table 2
Key parameters of the energy storage options and HVAC systems and tariff for the case study.

Parameter	Value	Parameter	Value
Battery energy storage		**TES tank with PCM**	
Battery type	Li-ion	PCM type	Salt-hydrate
Inverter efficiency	0.95	Thermal conductivity, W/m K	0.6
Max./min. State of charge	0.1/0.95	Density, kg/m[3]	1350
Max. discharge rate	1C	Specific heat capacity, kJ/kg °C	2.0
Charge rate	0.25C	Heat storage capacity, kJ/kg	155
Battery unit price[1], HKD/kWh	6100	Melting temperature, °C	4–8
Reference: [51]		Phase change temperature range, °C	2.0
BITES with PCM			
PCM type	Paraffin	Tube inner/outer diameter, m	0.018/0.016
Thermal conductivity, W/m K	0.2	Tube material	Copper
Density, kg/m[3]	900	Heat transfer fluid	Water
Specific heat capacity, kJ/kg °C	2000	Rated flow rate[2], kg/h kWh	180.6
Latent heat, kJ/kg	100	Charge temperature, °C	3.5
Melting temperature, °C	25	TES tank unit price[1], HKD/kWh	250
Phase change temperature range, °C	3.5	Reference: [40, 57, 58]	
BITES unit price[1], HKD/kWh	180	**Tariff**	
Reference: [57, 59, 60]		TOU on-peak hours[6]	09:00–21:00 Mon.-Sat.
HVAC system			
Zone temperature set point, °C	25.5	On-peak/off-peak electricity price, HKD /kWh	0.753/0.676
Chilled water set point, °C	7.0		
Rated chiller COP[3]	3.4/4.2	DR duration, hour	1.0/2.0/3.0
Rated cooling capacity[4], W/m[2]	220/250	DR event lead time, hour	0.5/4.0
Precooling set point, °C	23.5	DR incentive[7], HKD/kWh	20/8
Reset temperature tolerance, °C	0–2.5	IL threshold, kW	500
Rated fan/pump power for cooling distribution[5]	40% /30%	IL peak shaving threshold	20%
		IL incentive, HKD/kW month	100
		Reference: [53–55]	

[1] Unit price includes material cost and associated installation costs.
[2] Rated flow rate per unit storage capacity.
[3] Air-cooled/water-cooled chiller including cooling tower.
[4] Per air-conditioned area for buildings with regular/glass-curtain façade.
[5] Percentages of rated chiller power.
[6] All other hours are off-peak hours.
[7] 20/8 HKD/kWh for events with half-hour/four-hour lead time.

(i.e., PCM nodes) along the radial direction. The governing equations of the water nodes and the PCM nodes are presented in Eq. (2) and Eq. (3), respectively. The thermal resistance of the pipe wall is neglected. The heat loss through the outer surface of the TES tank to ambient is simplified as a source term of the PCM nodes, which is mainly determined by the PCM mean temperature and the ambient temperature [40]. The phase change process is modeled by the interrupted heating and cooling curves proposed by Delcroix et al. [46] to include the phase change hysteresis effect.

$$c_{p,w}m_w\frac{\partial T_{w,i}}{\partial t} = h_{w,conv}A_{tb}\left(T_{PCM,i}\big|_{r=r_{tb}} - T_{w,i}\right) - c_{p,w}\dot{m}_w\left(T_{w,out,i} - T_{w,in,i}\right) \tag{2}$$

$$2\pi r dr \Delta x \rho_{PCM} c_{p,PCM}\frac{\partial T_{PCM,i,j}}{\partial t} = 2\pi k_{PCM}\Delta x\left(\frac{T_{PCM,i,r+dr} - T_{PCM,i,r}}{ln\left(\frac{r+dr}{r}\right)} - \frac{T_{PCM,i,r} - T_{PCM,i,r-dr}}{ln\left(\frac{r}{r-dr}\right)}\right) - q_{loss} \tag{3}$$

where $c_{p,w}$ is the water specific heat capacity, m_w is the water mass in a node, $T_{w,i}$ is water temperature of the i^{th} node along the flow direction, t is time, $h_{w,conv}$ is the convection heat transfer coefficient on the water side, A_{tb} is the tube surface area of a node, $T_{PCM,i}\big|_{r=r_{tb}}$ is the PCM boundary temperature in contact to the pipe wall of the i^{th} node, $\dot{m}_w$ is the water mass flow rate, $T_{w,out,i}$ and $T_{w,in,i}$ are the outlet and inlet temperature of the water flow of the i^{th} node, $T_{PCM,i,j}$ is the PCM temperature of the (i^{th}, j^{th}) node where i indicates the water flow direction and j indicates the radius direction, r is the radius of the j^{th} PCM node, Δx is the length of a water flow node, ρ_{PCM} is the density of the PCM, $c_{p,PCM}$ is the specific heat capacity of the PCM which is determined by the heating and cooling curves, k_{PCM} is the thermal conductivity of the PCM, and q_{loss} is the source term of heat loss to ambient for a PCM node.

The BITES with PCM is modeled by a 1D finite-difference model developed by Kuznik et al. [47]. The PCM layer is modeled by the 1D finite-difference scheme and the other ceiling components are modeled by the lumped-capacitance method. The governing equation of the PCM nodes is presented in Eq. (4). Similar to the TES tank, the phase change process is modeled by the interrupted heating and cooling curves.

$$\rho_{PCM}c_{p,PCM}\frac{\partial T_{PCM,i}}{\partial t} = -\frac{\partial}{\partial x}\left(k_{PCM}\frac{\partial T_{PCM}}{\partial x}\right) \tag{4}$$

The HVAC system is modeled using components provided by the TRNSYS software and TESS library, mainly including the performance-map-based chiller models, i.e., Type 655 and Type 666, and the cooling tower model, i.e., Type 510. The chilled water return temperature is

Summary of DR events

Day	2	3	4	9	10	12	13	16	23	24	25	26	27	30
Lead time (hour)	0.5	0.5	4	0.5	4	4	0.5	4	0.5	0.5	4	0.5	4	0.5
Duration (hour)	1	2	3	2	3	1	2	1	1	3	2	3	2	1
Start time	14:00	15:00	12:00	13:00	12:00	14:00	14:00	15:00	13:00	15:00	15:00	14:00	15:00	14:00

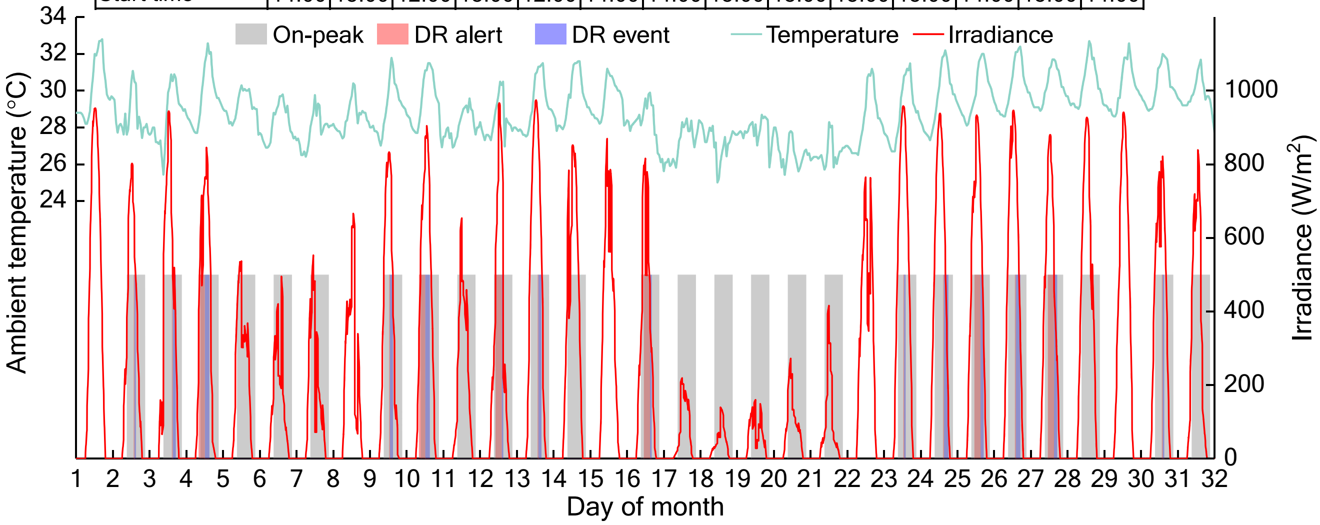


Fig. 6. TOU on-peak hours, DR events, and weather data in the simulation month.

calculated based on the energy balance between the cooling load and the chilled water. The pump and fan power consumptions are evaluated based on a cubic relationship between the flow rates and the power consumption [48].

3.3. Case study setup

A case study including 21 buildings located in a street block in Tsim Sha Tsui, Hong Kong was set up to evaluate the effectiveness of the data-driven surrogate optimization for optimal deployment of multi-energy storage at a building cluster level. The building cluster was selected to cover a wide range of building types and sizes. The key parameters of the building models are summarized in Table 1. The gross area, building type, façade type, WWR (windows-to-wall ratio), and building type were extracted from the Building Framework Spatial Data Theme of Hong Kong Geodata Store [49] and visual inspection of Google Map and Google Street View imagery. The HVAC type was determined based on the building type and the gross area, assuming residential buildings adopted split air-conditioners and commercial buildings adopted centralized air-conditioning systems. The temperature-reset tolerance was determined based on the building type and the façade type, assuming buildings with glass-curtain façade generally had lower tolerance due to high solar gain. Other building parameters were determined with reference to the local design code [50] and existing studies [51,52].

Table 2 presents the key parameters for modeling the energy storage systems, the HVAC systems, and the tariff for the case study. The unit price of the energy storage included costs for equipment and installation, and the initial costs were directly determined based on the unit price and the storage capacity. The parameters and costs of the battery, the TES tank with PCM, and the BITES with PCM, were determined based on previous studies as listed in Table 2.

The charge flow rate of a TES tank was proportional to the storage capacity and should not exceed the rated flow rate of the chiller(s). The flow rate was determined via trial tests, guaranteeing that the depleted TES tank could be fully charged in less than six hours (i.e., the regular charging duration) if the cooling power was sufficient. If charging was required when there was cooling requirement, the chilled water supply temperature will be reset to 3.5 °C to simultaneously provide cooling to the load side and to the TES tank, while the flow rate to the TES tank will be halved to avoid insufficient cooling supply to the load side. In the discharging mode, the discharging flow is regulated to maintain a large temperature difference between the return water and the PCM [40]. Considering the 3.5 °C charge temperature, the charging process was terminated when the PCM average temperature was less than 4.0 °C. The discharging process is suspended when the difference between the PCM average temperature and the return water temperature was less than 0.5 °C. The startup/shutdown of the TES included dead-band operations to avoid oscillation.

The individual building energy bill includes the electricity cost under the TOU tariff, the DR incentive, and the monthly interruptible load (IL) incentive. The electricity cost was evaluated based on the simulated grid import profile. The DR incentive and the interruptible load incentive were evaluated based on the grid import reduction of a building with energy storage, as compared to the building without using energy storage or demand response control strategy (i.e., reference case). The TOU tariff and demand response programs were mainly determined based on programs provided by a local major power supplier [53,54]. An interruptible load program for DR events was included to formulate a diversified scheme [55]. The higher incentive, i.e., 20 HKD/kWh, was provided for the DR events alerted half an hour ahead since it is generally more valuable for the grid to reduce power consumption on shorter notice [53,56].

The simulation was conducted for a month of July with frequent DR events, using typical meteorological year weather data of Hong Kong [61]. The simulation time step was five minutes. Fig. 6 presents the TOU on-peak hours and the durations of the DR alerts and DR events in the simulation month, and the ambient temperature and horizontal solar irradiance. Frequent DR events, i.e., 14 events and 27 h in total, were arranged in the simulation month, to evaluate the performance of the multi-energy storage under complex operating conditions. The DR events were arranged on weekdays with high peak temperatures. The DR events were arranged from 12:00 to 18:00 when the local grid was usually stressed. To emulate complex situations, three different durations, i.e., one hour, two hours, and three hours, and two different alert lead times, i.e., half an hour and four hours, were arranged for the DR

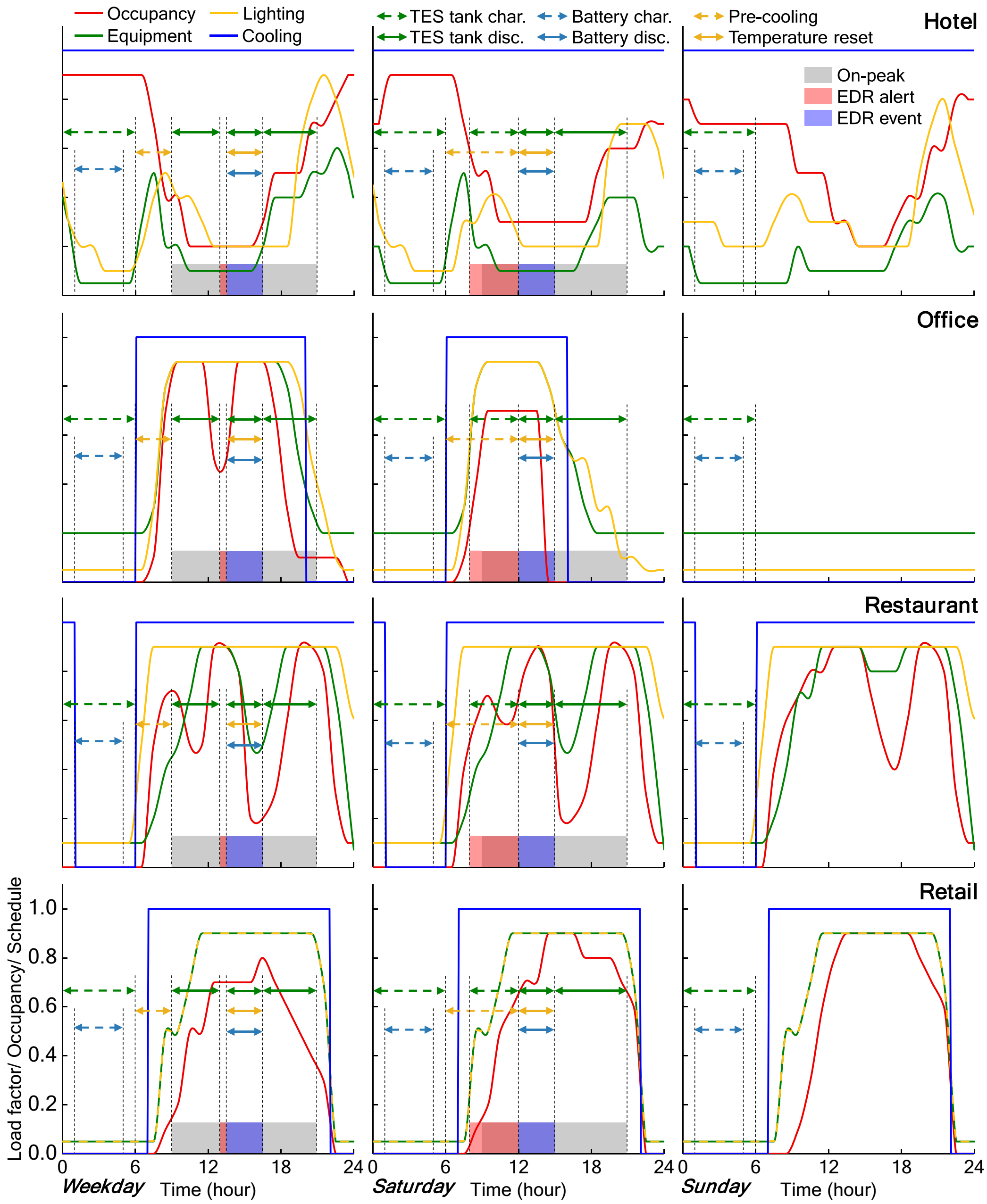


Fig. 7. Commercial building operating schedules and exemplary energy storage operation schedules.

events.

The operating schedules of the commercial buildings (i.e., hotel, office, restaurant, and retail) are presented in Fig. 7, which were based on the local design code [50]. To illustrate the multi-energy storage control strategies (as presented in Section 3.1), taking two DR events with different lead times and durations as examples, operating schedules of the energy storage systems are also presented in Fig. 7. The schedules of the residential buildings were formulated based on four typical schedules (i.e., F1 to F4) provided by a survey study of Hong Kong residents [62]. The north/south zones of the residential buildings #13-#18 adopted F1/F1, F1/F2, F1/F3, F3/F3, and F4/F4 schedules, respectively. The equipment and lighting load factors and occupancy were interpolated based on the provided hourly schedules, to accommodate the five-minute simulation time step.

When conducting the optimal deployment, the capacities of the battery, the capacities and phase change temperatures of the TES tanks, and the capacities of the BITES were considered the design parameters to be optimized (i.e., decision variables). The melting temperature of the BITES was excluded from the decision variables, since from 22 °C to 27 °C, its variation only slightly influenced the building cluster energy bill. The melting temperature of the BITES was 25 °C considering the 23.5 °C pre-cooling temperature. For the building cluster of concern, there were 72 decision variables in total.

To accelerate the search for the optimal solution and avoid unreasonable design, lower and upper bounds were defined for the decision variables of individual buildings. The lower and upper bounds for the storage capacities of batteries and TES tanks were adapted to individual buildings in correspondence to the respective electrical load and thermal load. The upper bounds of the battery capacities were equivalent to five times the average hourly equipment and lighting load of the respective buildings. Similarly, the upper bounds of the TES tank capacities were equivalent to five hours times the rated cooling capacities of the air-conditioning systems (e.g., a 1-kW rated cooling capacity corresponding to a 5-kWh upper bound). The lower bounds for the capacities of the

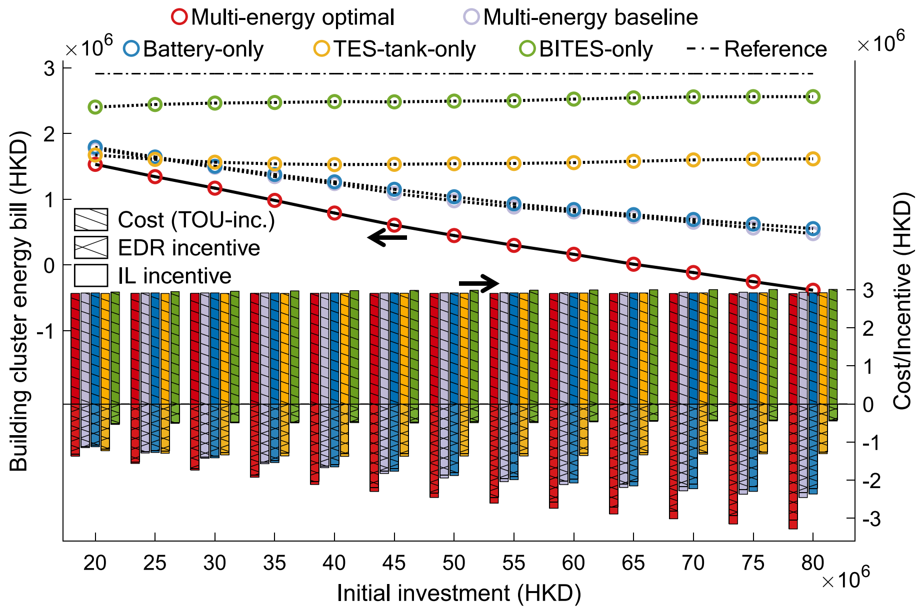


Fig. 8. Building cluster energy bills and bill breakdowns in the simulation month.

batteries and the TES tanks were 5% of the respective upper bounds to avoid unreasonable sizing. The lower and upper bounds for the PCM melting temperature of the TES tank was 4 °C to 8 °C based on the operating range of the chilled water. Since the BITES was applied to all air-conditioned areas, the capacity of the BITES was controlled by the PCM layer thickness which was from 0.005 m to 0.03 m, with reference to existing studies [63,64].

In the defined lower and upper bounds, the decision variable samples of the training data were formulated with 300 samples generated by the Latin hypercube sampling, and another 60 samples were formulated for the screened energy storage options. Extra 50 samples were generated for the validation of the surrogate models. By using the symbolic regression, the surrogate models achieved high accuracy in predicting the individual building energy bills. The R^2 of surrogated models were all above 0.98 and the mean absolute percentage error of the validation for predicting the building cluster energy bill was less than 4%.

To evaluate the performance of the optimal deployment plan and conduct detailed analysis, baseline cases of the energy storage deployment at a cluster level were formulated, including a multi-energy storage baseline case, and three baseline cases solely using one of the energy storage options, i.e., batter-only case, TES-tank-only case, and BITES-only case. The multi-energy baseline case allocated budgets for buildings proportionally to their total energy consumption and further allocated the budgets of an individual building at a ratio of 2:1:1 to the battery, the TES tank, and the BITES. The battery-only case, the TES-tank-only case, and the BITES-only case allocated the budgets proportionally to the equipment and lighting loads, the rated cooling capacities of air-conditioning systems, and the air-conditioned areas of the buildings, respectively. The grid-import reduction of all cases was evaluated by comparing them to the reference case that did not include any energy storage or demand response strategy.

4. Results and analysis

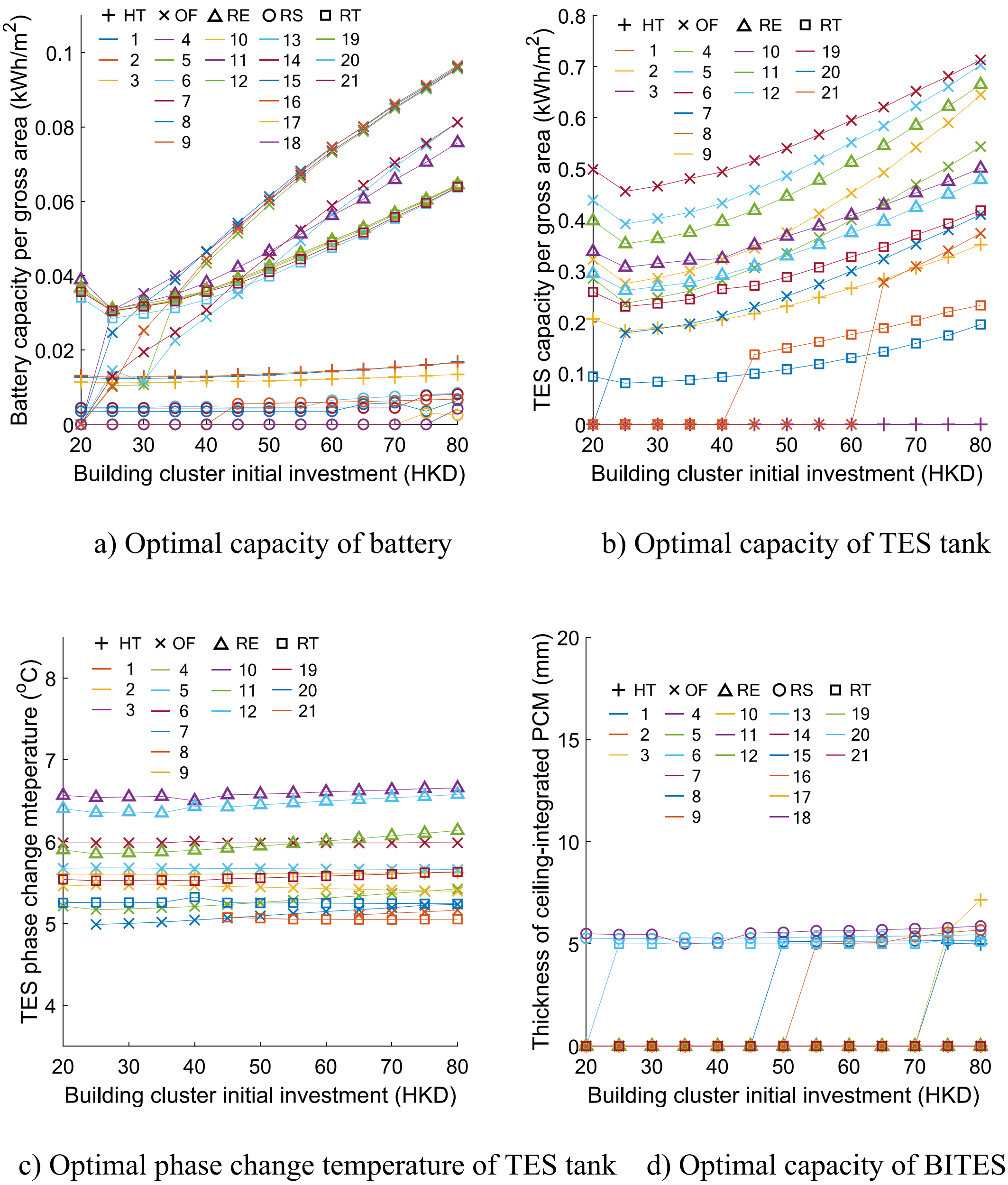
4.1. Optimization results

Fig. 8 presents the building cluster energy bills and bill breakdowns of the optimal cases and the baseline cases when the budget constraint ramped from 20 million HKD to 80 million HKD. The optimal deployment method reduced the building cluster energy bill by 8%–181%, as compared to the best of the baseline cases, and the reduction was more significant when the budget was higher. Taking the case without storage as the reference, the optimal deployment method increased the energy bill saving by 12%–36% as compared to the best of the baseline cases. The reduced energy bill was mostly contributed by the increasing incentives (mostly the DR incentive), while the change of electricity cost was insignificant due to the small difference between the on/off-peak rates (~10%). The optimal deployment increased the DR incentive by 12%–31%, as compared to the best of the baseline cases.

Thanks to the high discharge rate (i.e., 1C) of the battery, the battery-only cases achieved comparable performance to the multi-energy baseline cases and outperformed other baseline cases when the budget was more than 30 million HKD. The TES-tank-only case outperformed the battery-only case when the budget was less than 30 million HKD since the unit price of the TES tank was much less than the battery and it provided large storage capacity at low budgets. However, with the increase of the budget, the incentives were first increased and then decreased due to oversize issue. The BITES was the least cost-effective of the three options. The building cluster energy bill of the BITES-only case was even slightly increased with the increase of the budget. This was because the BITES was only suitable for a few buildings. Simply deploying BITES to all buildings would increase the HVAC energy cost while earning limited incentives.

Fig. 9 presents the optimal energy storage configurations and design parameters selected for individual buildings when the budget constraint ramped from 20 million HKD to 80 million HKD, where the above-zero capacities indicate the adoptions of corresponding energy storage options. In general, the optimal deployment plan included more energy storage options and larger storage capacities along with the budget increase (see Fig. 9a, b, and d), while the optimal phase change temperature of the TES tank was almost unaffected by the budget changes (see Fig. 9c). There was a clear trend that the optimal battery capacities of the hotel buildings and the residential buildings were only slightly increased with the budget increase, while those of the other buildings were substantially increased. The optimal TES tank capacities were much larger than the batteries, while the increase of the TES tank capacity (along with the budget increase) was much less significant. The number of buildings selecting the BITES increased from only two to eight when the budget increased from 20 million to 80 million, and the optimal capacities of the BITES were mostly close to the lower bound.

To further analyze the effectiveness of the optimal deployment method, Fig. 10 presents detailed contributions of the individual



a) Optimal capacity of battery

b) Optimal capacity of TES tank

c) Optimal phase change temperature of TES tank

d) Optimal capacity of BITES

Fig. 9. Optimal energy storage configurations and design parameters identified by the optimal deployment method (where HT, OF, RE, RS, and RT indicate hotel, office, restaurant, residential, and retail, respectively).

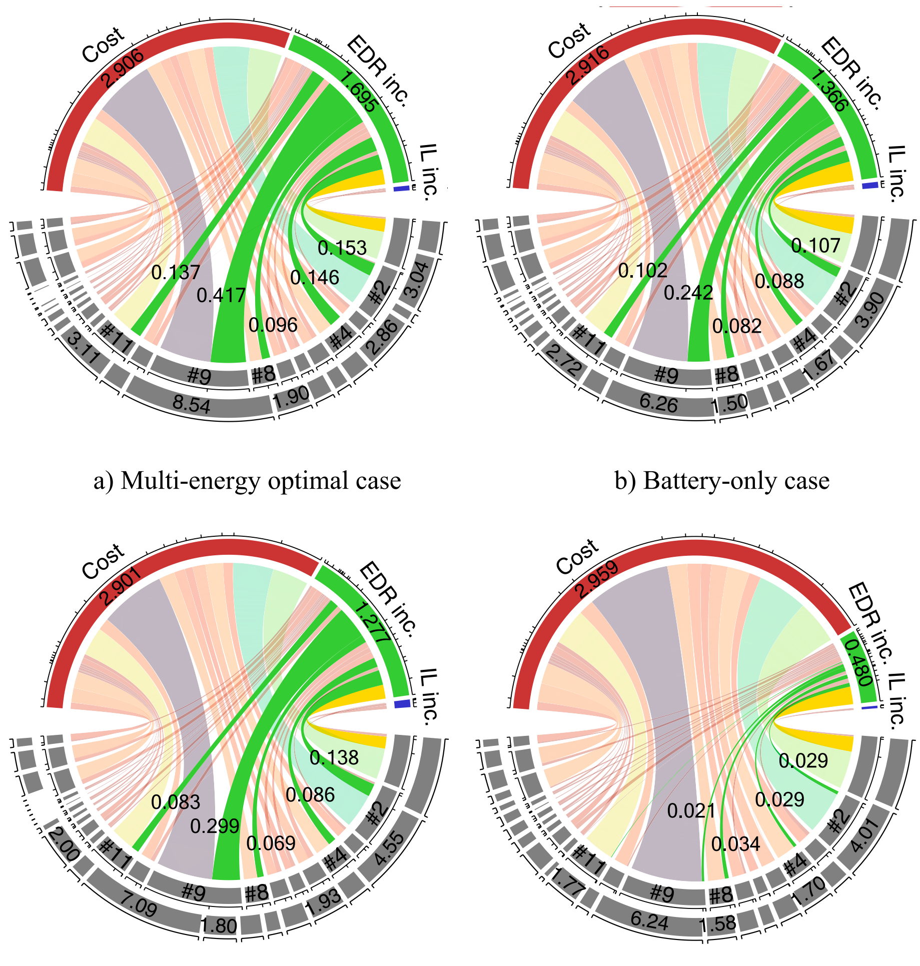
buildings to the building cluster energy bill, under the 30 million budget constraint. The increased DR incentive was largely contributed by buildings #2, #4, #8, #9, and #11, as highlighted in green. The optimal deployment method effectively identified the individual buildings with better potential to reduce grid import during DR events, and deployed multi-energy storage with optimal configurations and optimal design parameters. Therefore, cost-effectiveness can be achieved on a building cluster level. For instance, thanks to the optimal multi-energy configuration, i.e., 311kWh battery and 4584kWh TES tank, the initial investment of building #2 was reduced by 22%/33% as compared to the battery-only/TES-tank-only case, i.e., 640kWh/18,203kWh storage capacity, while its DR incentive was increased by 43%/11%. It is also noteworthy that building #1 achieved relatively high DR incentives in all cases (highlighted in yellow). This was mainly because building #1 was a hotel building with large thermal mass and 1.5 °C temperature reset tolerance, which significantly reduced the HVAC power during DR events via the demand response control strategies.

The optimal multi-energy storage configurations and design parameters were very different for individual buildings, resulting in highly diversified demand response performance among the buildings. Despite the different building gross areas, such diversity was mainly caused by two key factors and their joint effects: 1) Heterogeneity of multi-energy storages, i.e., different types of energy, charge/discharge rates, and unit costs; and 2) Diversified building energy use patterns. Detailed analysis for each energy storage option is as follows.

4.2. Detailed analysis

4.2.1. Battery

The battery energy storage provided high discharge power during DR events to reduce grid import, which makes it suitable for most buildings in the case study. However, in hotel buildings and residential buildings, there were temporal mismatches between the DR events and equipment and lighting loads. In the baseline cases without considering such



a) Multi-energy optimal case b) Battery-only case

c) TES-tank-only case d) BITES-only case

Fig. 10. Contributions of the individual buildings to the building cluster energy bill (where the inner circles indicate cost/incentive contributions, the outer semicircles indicate initial investments, and the unit of labels is million HKD) [65].

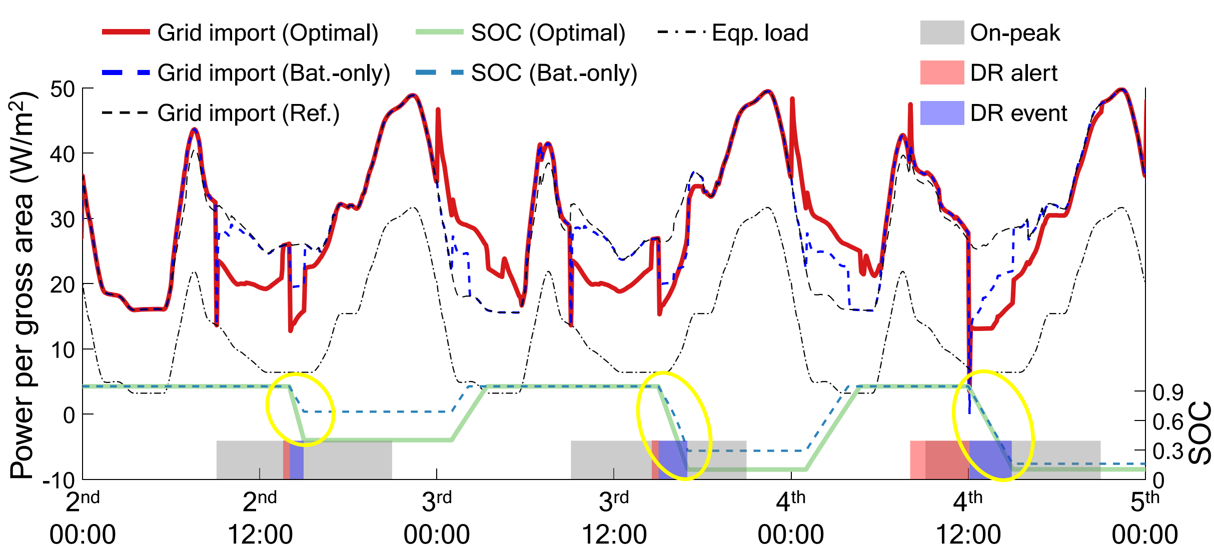


Fig. 11. Grid import and battery SOC of building #2 (30 million budget constraint).

temporal mismatches, the battery could be oversized, thereby resulting in substantial waste of the budget. Fig. 11 presents the grid import power and battery SOC of building #2 (i.e., a hotel) on the 2nd-4th simulation days of the optimal case and the battery-only case under the 30 million budget constraint. The equipment and lighting load during DR events was relatively low due to the energy use pattern of the

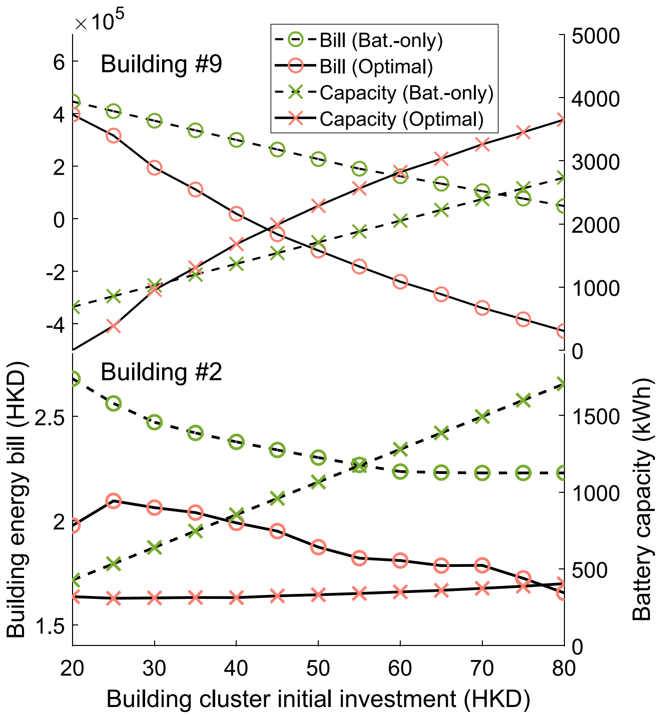


Fig. 12. Energy bill and battery capacity of buildings #2 and #9.

hotel. Therefore, the 640kWh battery of the battery-only case was always partially discharged (highlighted by the yellow circles), indicating that the battery was oversized, and the budget was wasted. The optimal case adopted a much smaller battery (i.e., 311kWh) and a 4584kWh TES tank, improving grid-import reduction during DR events while costing 22% less than the battery-only case.

Fig. 12 presents the energy bill and battery size of buildings #2 and #9 in the optimal case and the battery-only case, when the total budget increased from 20 million to 80 million. Unlike building #2 which was a hotel building with peak equipment and lighting load at night, building #9 was a large office building that peaked around noon. In the battery-only case, since the battery could be always fully discharged to earn the DR incentives, the energy bill of building #9 was reduced linearly with the increase of the battery capacity. In the optimal case, the optimal battery capacities were higher than those of the battery-only cases when the budget was more than 30 million HKD, to more effectively reduce the equipment and lighting load during DR events. On the other hand, increasing the battery capacity of building #2 could not effectively reduce the energy bill. Therefore, the optimal battery capacity was only slightly increased along with the budget increase. The reduced energy bill of building #2 (in the optimal case) was mainly contributed by the increased TES tank capacity.

4.2.2. TES tank with PCM

When compared to the battery, the TES tank with PCM provided a large storage capacity at a relatively low cost. However, the discharge rate, i.e., cooling power per storage capacity, of the TES tank was rather limited due to the low thermal conductivity of the PCM. The TES tank could consistently reduce grid import during on-peak hours (09:00 to 21:00), but might not achieve grid-import reduction as high as the battery during the DR events. Fig. 13 presents the grid import and TES tank average temperature of building #11 (i.e., a restaurant building with high cooling load) on the 2nd-4th simulation days of the optimal case, and the TES-tank-only case under the 30 million budget constraint. In the DR events, the grid-import reduction of the optimal case was always higher than the TES-tank-only case, which was mainly contributed by the battery. In the one-hour DR event on the 2nd day, the 351kWh battery and the 3887kWh TES tank of the optimal case provided 23.9kWh/m^2 and 9.8kWh/m^2 grid-import reduction, respectively, while the 8020kWh TES tank of the TES-tank-only case provided 15.5kWh/m^2 grid-import reduction. The large-size TES tank provided a greater amount of reduction during other on-peak hours, while the economic benefits were limited due to the small difference between the on/off-peak rates (~10%).

It is also noteworthy that the cooling power of the TES tank could be deteriorated when it was oversized (as compared to the chiller cooling capacity), due to insufficient precooling, i.e., charging. This is because an oversized TES tank could be only partially precooled during the off-peak hours, resulting in relatively high temperature of the storage media and reducing its heat transfer rate with the return chilled water (when the tank is discharged). The proposed method could optimally size the TES tank to avoid such effects, thereby effectively reducing chiller power during DR events. Fig. 14 presents the chiller power reduction and the TES tank average temperature of building #11 in the optimal case, and the TES-tank-only case with an oversized TES tank under the 80 million budget constraint. The optimally sized tank (i.e., 7118kWh) was more sufficiently cooled before the DR event, achieving a lower tank average temperature (highlighted by the red circle), as compared to the oversized tank (i.e., 21386kWh). The average temperature of the optimally sized TES tank was cooled to 3.7 °C, while the average temperature of the oversized tank was only reduced by ~1 °C to 6.3 °C. As a result, the optimally sized TES tank provided even higher instant chiller power reduction in the aftercoming DR event (highlighted by the blue circle). The insufficient precooling of the oversized TES tank was mainly due to the limited flow rate of the chiller which could not match the capacity of the tank. Therefore, the chiller could not fully charge the oversized TES tank during the provided hours.

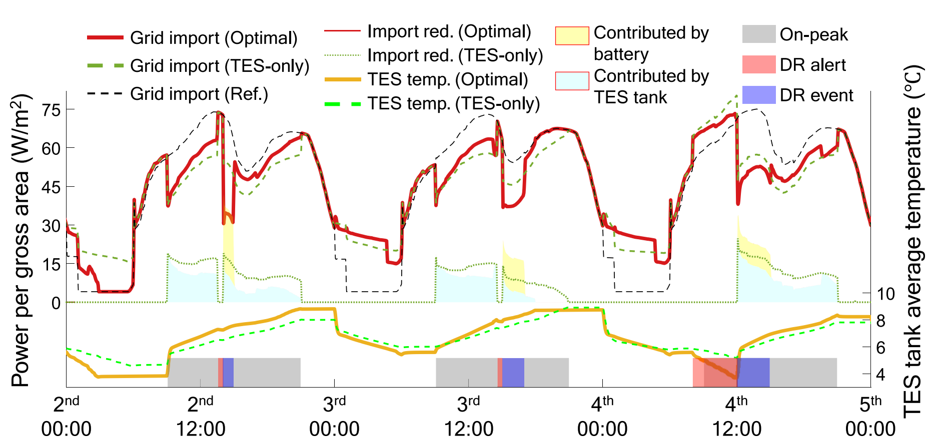


Fig. 13. Grid import and TES tank average temperature of building #11 (30 million budget constraint).

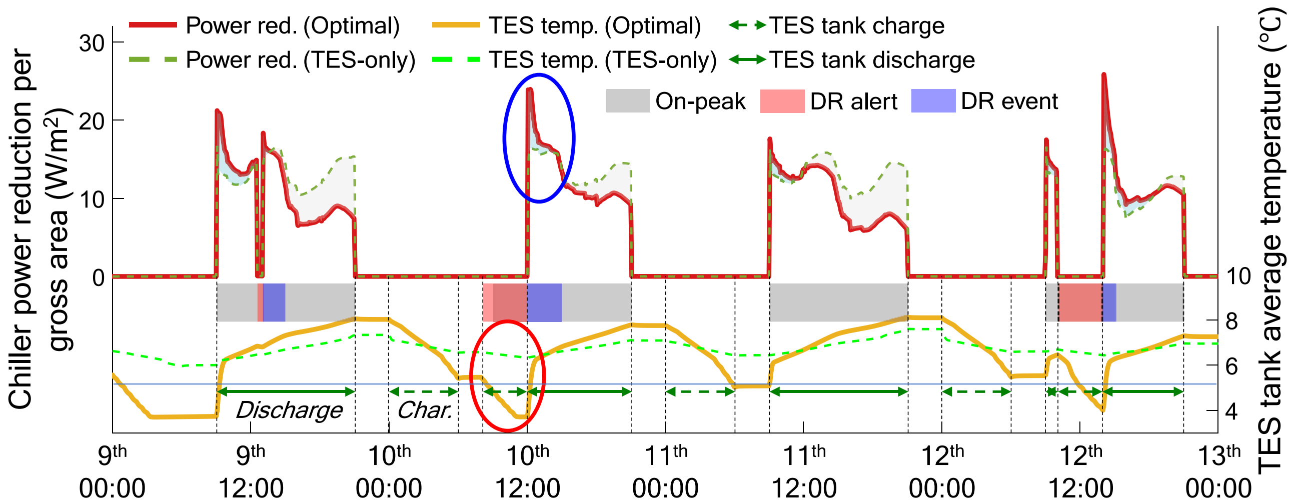
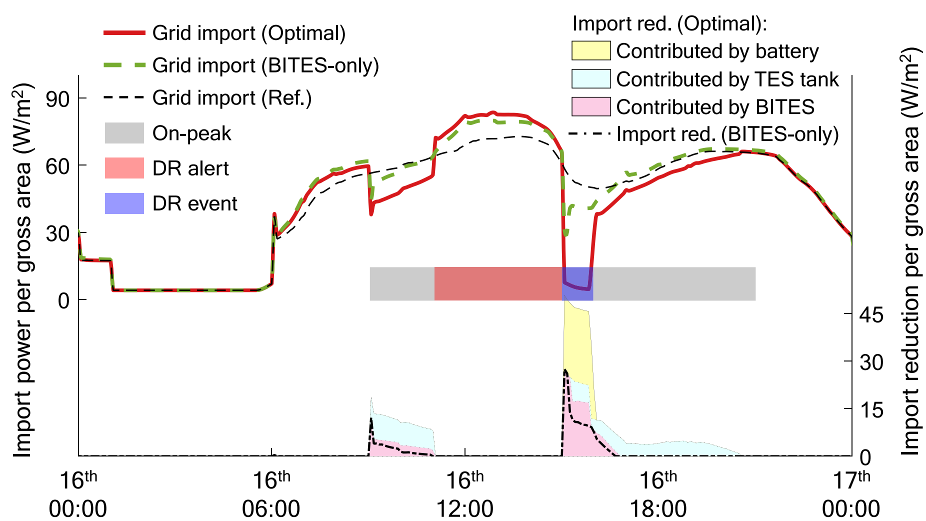
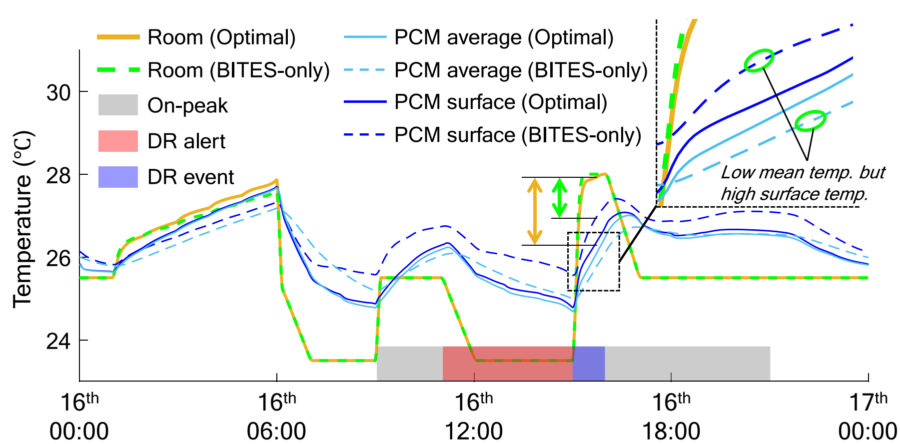


Fig. 14. Chiller power reduction and TES tank average temperature of building #11 (80 million budget constraint).



a) Grid import and breakdown of grid-import reduction.



b) PCM temperature and room temperature.

Fig. 15. Grid import and BITES PCM temperature of building #20 (80 million budget constraint).

4.2.3. BITES with PCM

The BITES was the least frequently used energy storage option in the case study. This was mainly caused by the poor heat transfer performance and the resulting low cooling power of the BITES. The BITES could only effectively reduce HVAC power when it was sufficiently precooled and the building had high temperature-reset tolerance.

Fig. 15 presents the grid import, breakdown of grid-import reduction, zone temperature, and PCM temperatures of building #20 in the optimal case (with 166kWh battery, 464kWh TES tank, and 5 mm-thick BITES) and the BITES-only case (with 52 mm-thick BITES), under the 30 million budget constraint. In the optimal case, the BITES provided comparable grid-import reduction (as the TES tank and the battery) during the on-peak hours (see Fig. 15a). The BITES provided higher grid-import reduction during the DR event than the other on-peak hours. This was because a 2.5 °C temperature reset was adopted by building #20 (see Fig. 15b), reducing the cooling load and facilitating the discharge of cooling energy by increasing the temperature difference between the BITES and indoor.

It is noteworthy that the BITES thickness of the optimal case was much less than that of the BITES-only case, while the optimally sized thinner BITES achieved higher grid-import reduction (see the contribution of BITES in Fig. 15a). This was because the thick PCM ceiling in the BITES-only case resulted in higher thermal resistance and deteriorated the heat transfer performance when discharged. Fig. 15b shows that the optimally sized BITES provided larger temperature difference between the surface (towards the indoor) of the PCM and the indoor air (highlighted by the arrows). The zoom-up in Fig. 15b shows that the thick PCM ceiling of the BITES-only case provided lower PCM mean temperature but higher PCM surface temperature as compared to the optimal case, due to the low thermal conductivity of the PCM. This indicated that the BITES must be properly sized to ensure that the cooling energy stored could be effectively discharged.

5. Conclusions

This study proposed a novel data-driven surrogate optimization for deploying heterogeneous multi-energy storage to improve demand response performance at a building cluster level. The proposed optimal deployment method adopted data-driven surrogate models to learn the demand response performance of individual buildings, and utilized an iterative optimization technique to identify the optimal configurations of multi-energy storage and optimal design parameters that minimized the energy bill at a cluster level. Three energy storage options including the battery, the TES tank with PCM, and the BITES with PCM were adopted. The effectiveness of the proposed method was verified by a building cluster including 21 buildings with diversified building types and HVAC configurations.

The optimal deployment method reduced the building cluster energy bill by 8%–181% when the budget for the multi-energy storage was 20 million-80 million, as compared to the best of baseline cases. This was mainly attributed to the DR incentives which were increased by 12%–31%. The proposed method effectively identified the buildings with higher potential to reduce grid import during DR events and the optimal configurations and sizing of the multi-energy storage for individual buildings. The pros and cons of the energy storage options were balanced by the deployment method to improve the demand response performance. The budget was cost-effectively allocated to individual buildings. Further analysis showed that the heterogeneous performance of the multi-energy storages (i.e., types of energy, charge/discharge rates, and unit costs) and the diversified building energy use patterns needed to be matched to achieve the optimal deployment. The battery provided a high discharge rate in a short period while its capacity was generally limited due to high unit cost. It was more suitable for buildings with higher equipment and lighting load during DR events. The TES tank could provide large storage capacity and consistent chiller power reduction during on-peak hours at a relatively low cost, while its discharge rate during DR events was limited. The BITES needed sufficient precooling and high temperature reset tolerance to effectively reduce HVAC power.

The data-driven surrogate optimization method for optimal deployment of multi-energy storage at a building cluster level can be a useful tool to guide the deployment of building energy storage systems at a large scale. The deployment of energy storage can facilitate buildings to engage in demand response and help improve the efficiency and reliability of grid operation. The developed method can be applied for both new planning and retrofitting scenarios and the applied building cluster could be a campus building cluster, an industry park, or a multi-use street block. This method can be helpful to various stakeholders such as property owners, energy performance contracting businesses, energy aggregators, and planning departments in local government. Although the current study has verified the effectiveness of the proposed data-driven surrogate optimization method, the generalization capability of the data-driven models remains uninvestigated. Also, the training data still needs to be generated from expert knowledge which is expensive to acquire. To address these limitations, we will further investigate surrogate modeling of building energy systems, to develop data-driven models with better generalization and to improve the usability of the proposed method.

CRediT authorship contribution statement

Haoshan Ren: Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Dian-ce Gao:** Writing – review & editing, Conceptualization. **Zhenjun Ma:** Writing – review & editing. **Sheng Zhang:** Writing – review & editing. **Yongjun Sun:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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